AI-Driven Optimization of Passenger Flow: Integrating Computer Vision, Machine Learning, and Simulation for Enhanced Efficiency and Revenue Generation

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Abstract**—** As Industry 4.0 continues to evolve rapidly, the digital transformation of airport operations has become essential for optimizing efficiency, improving passenger experience, increasing non-aeronautical revenue, and promoting sustainability. Traditional airport management approaches often struggle to balance efficient passenger flow with revenue generation, leading to congestion and missed commercial opportunities. This paper proposes an intelligent system that integrates Computer Vision (CV) for data acquisition, Machine Learning (ML) to predict movement patterns and spending behaviors, and Simulation to optimize passenger handling. Using a hybrid simulation framework that combines Agent-Based Modeling (ABM) and Discrete Event Simulation (DES), the system dynamically tests various boarding strategies, prioritizing boarding sequences based on commercial engagement and location proximity, to maximize revenue and maintain operational efficiency. This study shows how AI-driven decision-making, combined with simulation techniques, can enhance boarding efficiency and revenue generation, offering a scalable solution adaptable to various aviation environments.

Keywords—* *Industry 4.0, Airport Optimization, Computer Vision, Machine Learning, Simulation, Passenger Flow Management, Revenue Maximization, IoT, Artificial Intelligence, Agent-Based Modeling, Discrete Event Simulation.

I. Introduction

With the rise of Industry 4.0, digital transformations have become essential for optimizing aviation services while promoting sustainability. The aviation industry continuously evolves, from early gliders to fully automated aircrafts operating within complex airport ecosystems [1]. Managing airport systems requires addressing intricate decision-making processes that balance passenger satisfaction, revenue maximization, and environmental sustainability [2], [3].

Advancement in Artificial Intelligence (AI) and ML now enables strategic optimization of airport operations. Various intelligent systems have been proposed to enhance decision-making, leveraging ML techniques for automation and improved efficiency [4]. In particular, CV Technology facilitates accurate data collection, serving as a critical input for simulation and decision-making models.

One of the primary challenges airports face is balancing efficient passenger flow with non-aeronautical revenue generation, including earnings from retail, dining, and other commercial services. Traditional operational strategies often struggle to minimize congestion while maximizing commercial engagement, resulting in inefficiencies in both areas.

This paper presents a conceptual framework designed to ensure seamless passenger flow that integrates CV, ML, and Simulation to optimize passenger movement within airport terminals while maximizing non-aeronautical revenue. The approach begins with analyzing passenger behaviour to determine the optimal timing for boarding calls. By selectively dispatching passengers, the system aims to enhance revenue opportunities and reduce congestion at the gate.

The paper is structured as follows: Section II reviews existing literature on the role of technology and simulation methods in industrial systems. Section III outlines the proposed methodology, detailing the overall pipeline and the contributions of CV, ML, and Simulation in implementing the concept. Finally, Section IV presents conclusions and future perspectives.

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II. LITERATURE REVIEW

The rise of intelligent technologies in the 21st century has transformed industries by enabling automation, predictive analytics, and adaptive decision-making. Among these, Simulation, ML, and CV have been particularly ground-breaking, allowing machines to observe, predict and adapt with unprecedented accuracy.

Industries such as aviation, manufacturing, and urban planning increasingly rely on simulation frameworks to model outcomes, enhance resource utilization, and streamline complex operations [5]. Simulation serves as a powerful tool for modelling, testing, and optimizing processes in virtual environments before real-world implementation. For instance, event-driven simulations play a crucial role in airports by forecasting passenger movement, reducing congestion, and optimizing terminal layouts [6]. One notable study [7] introduced an individual-based stochastic model that simulates passenger behavior within airport terminals. By accounting for tactical decisions and route preferences, the model allows planners to evaluate the effects of infrastructure configurations and operational strategies on terminal efficiency. Simulating various operational scenarios, such as, peak-hour check-ins, security wait times, and boarding gate assignments, provides planners with actionable insights to reduce bottlenecks and improve the overall passenger experience [5], [7].

In manufacturing, advanced simulation techniques like ABM have introduced AI-driven, self-optimizing production lines that predict equipment failures, detect inefficiencies, and autonomously adjust workflow [8]. ABM, when integrated with Cyber-Physical Systems (CPS), the Internet of Things (IoT), and real-time AI processing, evolves into a dynamic, self-learning system capable of predicting outcomes and enabling autonomous decision-making [9]. Whether applied in aviation, urban development or industrial automation, simulation provides a crucial bridge between digital planning and real-world execution.

CV, meanwhile, has significantly advanced automation by allowing machines to interpret visual data with near-human accuracy. CV enables machines to process images, recognize patterns, and make intelligent decisions without human intervention [9]. In the context of airport systems, CV is increasingly utilized to enhance operational efficiency and safety. For instance, [10] propose a novel CV framework for airport-airside surveillance, employing Convolutional Neural Networks and camera calibration to support aircraft detection, tracking, push-back prediction, and maneuver monitoring. The system demonstrated high precision in object detection and provides accurate estimations of aircraft speed and distance, contributing to improved safety and operational efficiency in airport environments.

While CV enhances automation through image-based recognition and simulation allows the development of virtual models for optimizing complex systems, ML complements both by enabling data-driven decision making and real-time optimization [11]. ML algorithms continuously learn from data, allowing industries to move beyond traditional rule-based operations toward adaptive, intelligent eco-systems. Its applications range from minimizing energy consumption in manufacturing to predictive maintenance in industrial systems through smart retrofitting [12], [13]. In smart airports, ML has improved cybersecurity by detecting anomalies in IoT-based networks and preventing data breaches. Additionally, ML-powered biometric recognition systems have enhanced passenger flow automation, reducing wait times and improving security [11]. As industries evolve, ML will remain central to digital transformation, driving self-optimizing ecosystems that improve operational efficiency, resilience, and intelligence [13].

Further, ML enables data-driven decision-making and real-time optimization by learning from historical and streaming data. In a recent study, [14] introduced a hybrid deep learning architecture for accurate airport express passenger flow forecasting. Their work underscores the effectiveness of combining spatial and temporal features for intelligent transportation prediction. Similarly, [15] explored hybrid artificial neural networks integrated with optimization algorithms such as parliamentary optimization and intelligent water drops, resulting in improved performance in passenger flow prediction. Beyond forecasting, [16] developed an AI-based simulation framework for managing airport terminal crowds, enabling scenario testing for staff deployment and infrastructure changes to reduce wait times and improve user experience.

Together, simulation, CV, and ML form a synergistic triad. Simulation provides a risk-free environment to model, test and optimize strategies; CV provides real-time perception for automation, safety, and quality control; and ML empowers systems with self-learning capabilities and adapt decision making based on real-time feedback [11], [13].

One key application area where this convergence is especially valuable is airport passenger flow optimization, a critical component of efficient operations. Various boarding strategies, including 'back-to-front', 'outside-in', and 'random' boarding, have been developed to minimize boarding time and enhance passenger satisfaction. Each method has its advantages and limitations [17]. For instance, the 'back-to-front' method is simple but often results in longer boarding times due to aisle congestion while the 'outside-in' strategy, which boards window seats first, aims to reduce seat conflicts but can be difficult to implement in practice. Passenger handling processes at airport terminals also play a significant role in overall efficiency. Agent-based models and discrete-event simulations have been employed to model passenger movements and identify bottlenecks in terminal operations [18]. These models provide valuable insights into the impact of passenger grouping, baggage handling, and security checks on terminal throughput.

While previous literature extensively explored CV, ML, and simulation independently, few have proposed a tightly integrated approach where these technologies form a sequential and interdependent pipeline for operational optimization.

This paper bridges that gap. By integrating real-time data acquisition (via CV), predictive modeling (via ML), and scenario optimization (via simulation), the proposed framework enables adaptive, real-time passenger flow management while maximizing non-aeronautical revenue. This framework reimagines traditional, static airport operations as a data-driven, self-learning ecosystem. The convergence of these technologies not only streamlines boarding and enhance passenger satisfaction, but also exploit commercial engagement opportunities—an integrated proactive strategy essential for modern airport environments and currently underrepresented in existing literature.

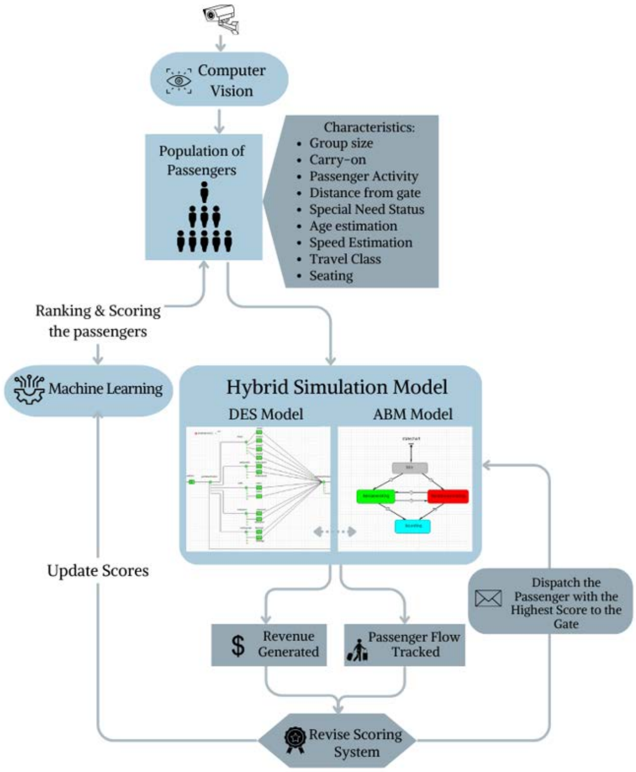

Figure 1: Overall System Pipeline

III. METHODOLOGY

The proposed system consists of three interconnected modules—CV, ML, and Simulation—that form a continuous feedback loop to optimize airport operations, as illustrated in Figure 1. The process begins with the CV module, which leverages CCTV footage to extract key passenger attributes in real time, including group size, carry-on quantity, passenger activity, distance from the gate, special needs status, estimated age, and estimated walking speed. By continuously monitoring passenger behavior, the CV module will transform raw video data into structured insights.

This structured data is then processed by the ML module, which employs two XGBoost Regression models to analyse passenger behaviour. The choice of model is based on literature highlighting its superior ability for generalization and optimization efficiency [**19**]. The first regression model predicts passenger boarding response time based on factors such as distance from the gate, group size, and historical response patterns. The second model estimates non-aeronautical revenue potential by analysing spending patterns and demographic attributes. Each parameter is assigned a weight based on its impact, enabling a predictive layer that dynamically optimizes airport operations by adjusting boarding sequences and maximizing revenue generation.

The Simulation module integrates ABM and DES to replicate passenger movements within the terminal. This module is currently used to train the system by simulating passenger flow between 5 different activities: shops, restaurants, cafés, restrooms, and waiting areas, as well as to test various boarding strategies. Data from CV are stored in Excel files, which are then imported into AnyLogic to create the population of agents with their parameters. Excel is chosen for ease of visualization and compatibility with analysis tools during prototyping. An SQL database with batch processing support and indexing will be needed to manage large-scale simulation data more efficiently in the full-scale implementation.

At predetermined time intervals, passengers are ranked using a scoring function incorporating the ML-derived parameter weights. The passenger with the highest score is prioritized for boarding and directed to the gate. This iterative process continues until all passengers have boarded.

Throughout the simulation, key performance indicators (KPIs) such as revenue generated and queue time are measured. The parameter weights in the scoring function are evaluated based on these outcomes and used to retrain the ML model, refining the assigned weights before reintegrating them into the simulation. This iterative process continues until all passengers have boarded.

For real-world implementation, passengers would be encouraged to download an airport app by offering benefits such as discounts or Wi-Fi access. Through this app, they would receive personalized notifications prompting them to proceed to the gate at the optimal time. Additionally, facial recognition would verify each passenger's identity at the security checkpoint before entering the duty-free areas, ensuring seamless integration with the CV system. This holistic approach is expected to enhance operational efficiency, minimize boarding delays, and maximize commercial engagement.

A. Computer Vision Module

As illustrated in Figure 2, the proposed CV system architecture consists of three primary layers: Camera Layer, Core Processing Layer, and Data Layer, which work in harmony to enable efficient passenger flow analysis in airport environments.

1) Camera Layer

The Camera Layer serves as the foundation of the data acquisition pipeline, leveraging existing CCTV infrastructure within airports to minimize implementation costs. This approach eliminates the need for additional hardware, reducing both capital expenditure and maintenance costs while ensuring seamless integration with existing airport security systems. Two complementary approaches to video data acquisition address distinct operational requirements: real-time streaming for immediate analysis and offline acquisition for historical analysis and model training.

Real-time streaming enables instant passenger movement analysis by continuously capturing and processing video streaming memory. The system ensures reliable low latency video transmission through a robust streaming protocol. This streaming process works by establishing direct connections with the cameras, processing frames sequentially, and feeding the processed data directly into the analysis pipeline without intermediate storage. In contrast, the offline acquisition approach systematically archives footage for later analysis. The stored footage will serve multiple purposes: enabling detailed historical analysis, supporting pattern recognition over extended time periods, and providing training data for improving the implemented ML models. This approach integrates with Network Video Recording (NVR) systems, allowing efficient historical footage retrieval.

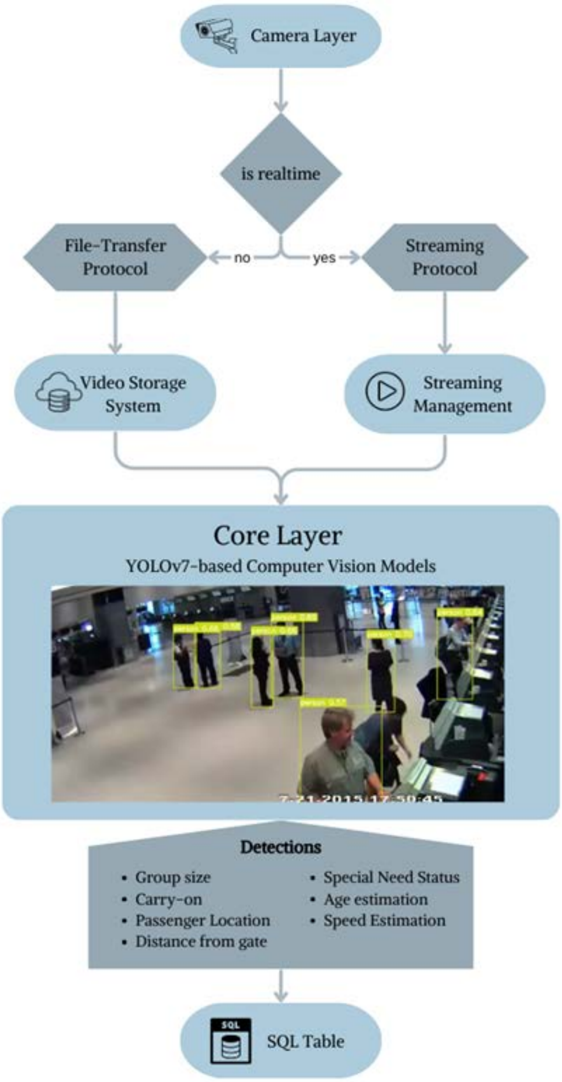

Figure 2: Computer Vision System Architecture.

2) Core Processing Layer

The Core Processing Layer implements a hierarchical multi-stage pipeline for video analysis and passenger profiling. Deployed on edge computing devices within the airport infrastructure and strategically positioned near CCTV camera clusters, the system minimizes latency and reduces bandwidth requirements. These edge devices process video streams locally but also manage bidirectional communication with the Data Layer for profile storage and the Camera Layer for stream management, creating a distributed yet cohesive processing network throughout the airport.

The Stream Management Service handles video pre-processing, frame extraction and synchronization, and integration with OpenCV for video manipulation. The AI Model Pipeline employs a transfer learning approach using YOLOv7 for object detection and tracking [**20**]. Key functionalities of the model include:

- **Person detection and tracking** using the CCTV Indoor Person Detection dataset and YOLOv7 with DeepSORT tracking integration [21].
- Age estimation using the Age Detection Dataset, categorizing passengers into ten age groups [22].
- Carry-on detection using the Luggage Dataset to differentiate between hard and soft case luggage [23].
- Special needs detection using the Detect Disabled and Old Person dataset to identify wheelchairs, canes, and elderly individuals [**24**].

Parameter extraction consists of two main components:

- **The movement component**: calculates speed using frame-to-frame displacement analysis, tracks location through camera position mapping, and estimates distances using camera-to-gate mapping algorithms.
- **The revenue attribution component**: implements zone-based mapping with expected revenue assignments (e.g., $20 for Shop 1, $50 for Restaurant 2, etc.), utilizing time-weighted accumulation and location-based analytics for accurate revenue tracking.

3) Data Layer

The Data Layer implements a temporal-relational data model optimized for both real-time updates and historical analysis. At its core is a structured SQL database that organizes CV detections and core processing outputs into a comprehensive frame-based schema. Each row in the primary tracking table represents a specific frame timestamp, while columns capture extracted passenger attributes such as counts, age distributions, carry-on luggage presence, special needs status, and movement parameters such as speed and location coordinates.

This structured approach enables detailed analysis of passenger behaviour patterns and facilitates ML applications. The frame-based rows provide temporal continuity in tracking, while feature-based columns allow for detailed profiling of passenger characteristics and their relationships to the KPIs. For instance, the database structure allows correlating passenger attributes like age, group size, and special needs status with movement patterns and spending behaviour in different airport zones.

The database supports the ML pipeline by analysing feature importance in relation to movement speed and aeronautical revenue. By maintaining a consistent, queryable dataset, the predictive model can be efficiently trained, updated, and utilized for real-time decision support.

B. Machine Learning Module

This module consists of two XGBoost Regression models, one for predicting passenger boarding response time and the other for estimating non-aeronautical revenue potential.

1) Data Sources

The ML module operates on a dual-input stream architecture. The primary input stream is sourced from the CV module, which provides real-time passenger data. The secondary input stream consists of real airport data during the training phase capturing boarding response times and spending behaviors. Due to current data access limitations, the Simulation Module is currently used to generate synthetic data for training and validation.

2) ML Models

The ML Module consists of the following regression models:

- **Boarding response time prediction model** processes historical response time vectors along with features such as normalized distance from the gate, age distribution, group size, special needs binary indicators, and carry-on quantity. The model outputs an estimated boarding response time (delay from when the boarding notification was sent to the passenger to when the passenger arrived at the gate).
- **Non--aeronautical revenue prediction model** analyzes historical spending patterns alongside the same input features, to estimate revenue potential.

By continuously refining the predictive models through simulation feedback, the system enhances operational efficiency, reduces boarding delays, and maximizes revenue generation.

3) Feature Evaluation and Impact Analysis

An initial correlation analysis was conducted on a synthetic dataset to evaluate the strength of relationships between passenger parameters and delay time in response to boarding. The synthetic dataset included behavioral and demographic parameters such as age, group size, walking speed, congestion level, distance to gate, and time of notification received. The model used was the XGBoost Regressor to predict the boarding response time of the passenger. The purpose of this experiment is a proof of concept of how ML will refine the weights in the scoring functions in the simulation model described later in Section III.C.

Figure 3 shows the results obtained where time-based features, such as Time of Notification Received, Estimated Arrival to Gate, and Actual Arrival to Gate exhibited the strongest correlations with the target, validating their critical influence on boarding delays.

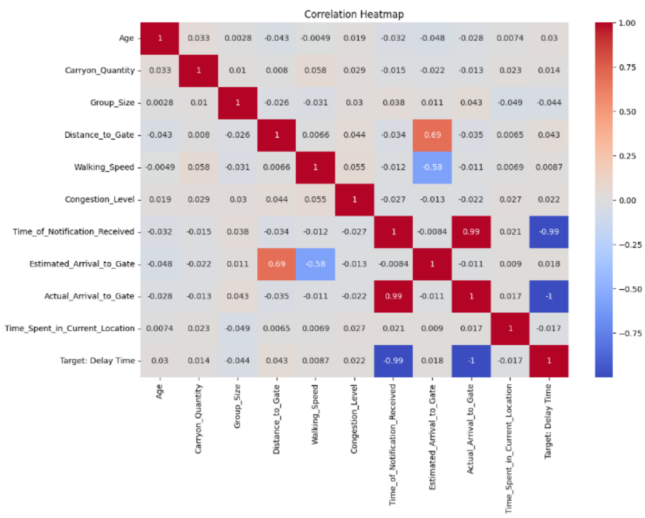

Figure 3: Correlation Heatmap of Passenger Parameters with Target Delay

These correlation results will also be operationalized by serving as weighting factors in the decision-making process for dynamically triggering boarding notifications, thereby enabling a personalized and adaptive passenger management strategy.

It is important to note that these findings are based on synthetic data and will be progressively refined through iterative simulation runs, which continuously update the ML model as more realistic behavior patterns are modeled. In the long term, as real-world operational data becomes available, the ML models will be retrained and validated against actual airport data, allowing the system to transition away from simulation dependency toward fully data-driven optimization.

C. Simulation Module

The simulation model creates a virtual testing environment to evaluate and optimize airport operations before real-world implementation. By replicating passenger flow, queue dynamics, and commercial interactions, it enables risk-free scenario testing to identify bottlenecks, determine the most efficient boarding strategies, reduce gate congestion, and maximize non-aeronautical revenue without disrupting actual operations. Using ABM and DES in AnyLogic, the model analyses passenger behaviour and boarding procedures under various conditions. Once received, CV data, including passenger demographics and movement speed, will be integrated into the simulation to enhance accuracy and validate weight assignments to key parameters. Additionally, the simulation module will initially act as a proxy data generator for the ML model, enabling the refinement of the delay function that estimates the optimal boarding time after a passenger receives a notification. As the ML model matures through exposure to real data, reliance on simulation will diminish.

1) Simulation Model Development

a) Agent-Based Passenger Representation

To establish a proof of concept and simulate real-world passenger movement, a synthetic dataset of 240 agents was generated using probability distributions from [25] and [26] ensuring a realistic representation of passenger characteristics as outlined in Section III.A. Each agent's attributes influence interactions within the terminal, affecting movement patterns, dwell time in commercial areas, and responsiveness to boarding notifications. The terminal layout is digitally recreated in AnyLogic as shown in Figure 4, inspired by a local airport and incorporating key zones such as security checkpoints, duty-free retail stores, cafés, restaurants, waiting areas, and lounges. While the initial configuration is randomized, the framework is designed to be adaptable to any airport layout, reinforcing the scalability of this approach.

b) Hybrid Simulation Approach: ABM and DES

The simulation model in AnyLogic combines ABM and DES to capture both individual passenger behaviour and overall airport flow. In the ABM component, passengers are modelled as autonomous agents with predefined parameters. The DES model uses the Process Modelling Library (PML) and the Pedestrian Library in AnyLogic, to simulate passenger movement within the duty-free area, modelling walking patterns, congestion effects, and commercial engagement. Passenger movement between activities—such as visiting a shop or café—is determined by a probability matrix imported from ML, which considers each passenger's profile defined by its parameters' values and current location to assess the likelihood of transitioning to the next activity.

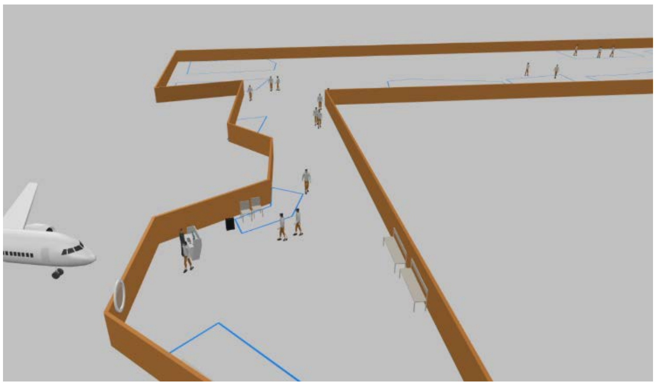

Figure 4: Demonstration of Passengers on AnyLogic

Passengers are dispatched to the gate using a cyclic event that periodically selects the top-scoring passenger based on a defined scoring function. The selected passenger is redirected from their current activity using the Cancel outport of the "pedwait" block and routed toward the boarding gate. Functions and visualizations are included to measure congestion levels and revenue generation, allowing for a continuous evaluation of the system's performance.

2) Dynamic Passenger Scoring and Boarding Optimization

To optimize boarding procedures, a dynamic passenger scoring system calculates a final boarding priority score (Z) by balancing boarding efficiency score (X) with revenue generation score (Y). The boarding efficiency score (X) increases as a passenger takes more time to arrive to the gate after receiving the boarding message, while the revenue generation score (Y) reflects the passenger's potential to generate non-aeronautical revenue (NAR) within the terminal.

This scoring system incorporates both fixed parameters, such as age, gender, group size, luggage size, special needs status, travel class, and seat location, and dynamic parameters, such as current location and activity. The fixed components of the scores X and Y are calculated using category-specific weights and parameter importance values imported from ML into the simulation as illustrated in Table 1 using the following equations.

$$X_{Fixed} = \sum_{Fixed\ Parameters} (Normalized\ Value\ of\ Category \times Boarding\ efficiency\ Weight\ of\ Parameter)$$

$$Y_{Fixed} = \sum_{Fixed\ Parameter} (Normalized\ Value\ of\ Category \times Revenue\ generation\ Weight\ of\ Parameter)$$

$$Z_{Fixed} = X_{Fixed} - Y_{Fixed}$$

Note that Table 1 presents a portion of the full dataset. Parameters such as seating assignment and travel group size are included in the complete dataset but are not shown here for brevity. The fixed component of the score Z, Z_{Fixed}, is the difference between X_{Fixed} and Y_{Fixed}.

The dynamic components of the boarding efficiency score, X, and the NAR score, Y, are calculated in real time within the AnyLogic simulation. These scores adjust dynamically based on the passenger's current location and activity. For example, revenue is incremented as follows: +\$50 for shopping, +\$20 for dining, +\$10 for cafés. Distance to the gate is recalculated every second using a cyclic event.

The final score is updated as follows where the weights are a tunable parameter.

$$Z_{Final} = Z_{Fixed} + (Normalized\ value\ of\ distance\ from\ gate\_category \times Boarding\ efficiency\ weight\ of\ parameter) - (Normalized\ value\ of\ activity\_category \times Revenue\ generation\ weight\ of\ parameter)$$

This combined scoring approach enables adaptive real-time decision-making that balances boarding efficiency with commercial revenue optimization. Higher-scoring passengers receive early boarding notifications, ensuring an optimal balance between maximizing non-aeronautical revenue and maintaining smooth passenger flow.

The highest-scored passengers are directed to the gate with an estimated arrival time. Their actual arrival times are recorded and fed into the ML module to refine predictions by analyzing delays linked to passenger activities and locations.

This feedback loop continuously updates the simulation, optimizing response times and boarding efficiency as illustrated in Figure 5.

Table 1: Fixed X-Y scores computation

		Boarding Efficiency			Revenue Generation			Fixed Score
Parameter	Category	Normalized Value of Category	Weight of Parameter	Fixed X Score	Normalized Value of Category	Weight of Parameter	Fixed Y Score	Z
Age	Young (<35)	0.15	0.175	0.02625	0.25	0.25	0.0625	-0.03625
	Old (>45)	0.7		0.1225	0.25		0.0625	0.06
	Adult [35 - 44]	0.15		0.02625	0.5		0.125	-0.09875
Gender	Female	0.6	0.075	0.045	0.7	0.3	0.21	-0.165
	Male	0.4		0.03	0.3		0.09	-0.06
Special Needs	Challenged	1	0.25	0.25	0.2	0.1	0.02	0.23
	Abled	0		0	0.8		0.08	-0.08
Traveling Class	Business	0.2	0.1	0.02	0.7	0.15	0.105	-0.085
	Economy	0.8		0.08	0.3		0.045	0.035
Carry-On Luggage	0 Bags	0.2	0.075	0.015	0.5	0.075	0.0375	-0.0225
	1 Bag	0.8		0.06	0.5		0.0375	0.0225

3) *Experimentation with boarding strategies*

The simulation module enables experimentation with different boarding strategies and scoring approaches to optimize both operational efficiency and NAR generation. Example of boarding strategies include:

- Dispatching passengers based on boarding efficiency alone (X score),
- Dispatching passengers based on potential NAR generation alone (Y score)
- Dispatching passengers based on a combination of both scores (Z score)
- Dispatching according to a weighted probability, based on the likelihood of engaging with additional NAR activities.

This flexibility allows for a more realistic and adaptive optimization of passenger flow. The simulation outputs key metrics such as the total revenue generated, maximum and average waiting time per passenger at the gate, and total time needed to process all passengers. These metrics inform decision making, train and iteratively refine the ML-based passenger weights in the scoring function. In the long run, as the ML model matures, the simulation will no longer be needed, and the ML component alone will drive the decision-making.

Additionally, the simulation evaluates the impact of delays between when a passenger receives a boarding instruction and the time they initiate movement, capturing real-time responsiveness. By continuously analysing passenger activity and refining the scoring system, the algorithm adapts in real time, ensuring the recommended boarding order evolves based on current behaviours. Through repeated simulations across multiple flights, the system fine-tunes parameters' weights, iteratively improving boarding efficiency through data-driven adjustments.

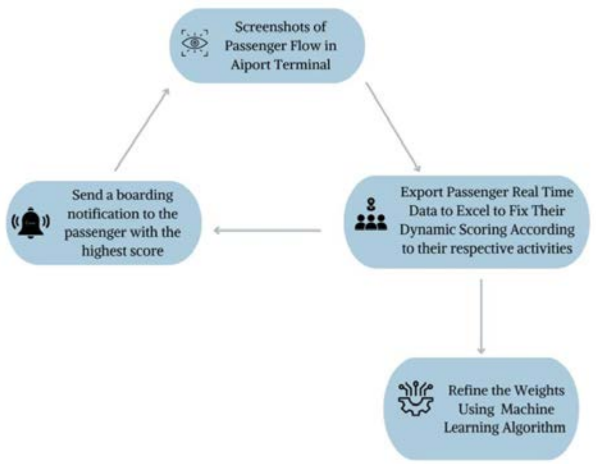

Figure 5: Simulation Model Iterative Process

IV. Passenger Privacy Concerns

Recognizing the sensitivity of CCTV-based passenger recognition, the system will include the following safeguards that align with GDPR principles and industry best practices for CV systems in public infrastructure:

Data minimization: This system is explicitly designed to infer non-identifiable, high-level attributes such as estimated age, group size, and baggage possession. No facial recognition or individual tracking is employed, which is in line with data minimization principles aimed at preventing the misuse of surveillance data [27].

Data retention policy: To ensure compliance with GDPR and privacy principles, the system implements strict data retention and on-device computing measures. All passenger data is processed in real-time and immediately discarded after use. No personal or identifiable data is retained beyond the immediate processing period.

On-Device processing: All data processing occurs directly on local devices (e.g., edge computing systems), meaning no sensitive information is sent to external servers for analysis or storage. This approach limits the risk of unauthorized access and minimizes data exposure.

Infrastructure and regulatory alignment: This system is designed to operate on top of the existing CCTV infrastructure deployed in airports, and thus adheres to the same regulatory and privacy policies already in place for these systems. This ensures that no new or unauthorized data acquisition mechanisms are introduced.

V. Conclusion and Future Outlooks

This paper proposes digitizing airport management systems through CV, ML, and Simulation modules. Existing literature was discussed on how different industrial practices tend to utilize AI technology, enhancing their work purpose and realizing new goals.

The overall pipeline was detailed, explaining the interconnection that exists between all three modules and how they contribute to a well-designed decision-making system rather than just depending on one module. The output of the CV architecture served as inputs for both ML and Simulation modules, with the output of the former also being an input for the latter.

While optimizing passenger flow management is just a glimpse of the endless operations taking place at the airport, it is possible to expand the concept of leveraging AI and Simulation modules to promote sustainable practices within the airport. To illustrate, IoT can be integrated to assist in designing processes that aim at detecting unsustainable practices and tackling them with the appropriate approach.

The developed framework holds great potential for successful integration into intelligent airport systems, and there are several exciting opportunities for future enhancements. A potential collaboration with a local airport to receive its CCTV footage is still under negotiation, and once realized, it will provide real-world data to refine and optimize the CV architecture. Although cross-domain applications were used as a proof of concept, they lay the foundation for the pipeline to reach its full potential in the future.

Finally, investing in advanced computational resources, such as high-memory GPUs and large-scale storage systems, will support the model's training and data management needs. These enhancements will ensure that the proposed framework evolves into a fully integrated, intelligent airport management system capable of addressing the complexities of modern aviation operations.

Match and Contribution

This paper aligns with the ICE IEEE/ITMC 2025 conference's theme of "AI-Enhanced Data-Driven Decision Making" by addressing the transformative role of AI and ML in revolutionizing airport operations. As the conference emphasizes AI's ability to optimize operations, enhance product development, and create new efficiencies in engineering and manufacturing, the paper directly contributes to this discussion by proposing an intelligent system that integrates CV, ML, and simulation techniques to optimize passenger flow and maximize non-aeronautical revenue in airport terminals. This approach not only enhances operational efficiency but also addresses key challenges in balancing efficient passenger management with revenue generation—critical components of digital transformation in aviation.

Acknowledgment

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